

Skills Summary

Signed Rational Numbers: Subtraction & Sign Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Subtracting signed numbers

Rule: $a - b = a + (-b)$ — subtracting means *adding the opposite*.

Same signs: add the absolute values and keep the sign. Different signs: subtract the absolute values and keep the sign of the number farther from zero.

Example: Find $-6 - 9$

Step 1. We are subtracting a positive number from a negative one, so rewrite the subtraction as adding the opposite and use the addition sign rules.

Step 2. $-6 - 9 = -6 + (-9)$. Same signs: add $6 + 9 = 15$ and keep the negative sign.
 $\Rightarrow -6 - 9 = -15$

Try it: Find $-4 - 9$

check: -13

(!) Watch out: the order matters — $a - b$ and $b - a$ are opposites. Never swap the two numbers just because one is negative.

Skill 2 — Subtracting a negative

Rule: $a - (-b) = a + b$ — two minus signs in a row make a plus.

Example: Find $-3 - (-10)$

Step 1. The number being subtracted is negative, so change the double negative into addition before computing anything.

Step 2. $-3 - (-10) = -3 + 10$. Different signs: subtract $10 - 3 = 7$ and keep the plus sign, because 10 is farther from zero.

$\Rightarrow -3 - (-10) = 7$

Try it: Find $-8 - (-3)$

check: -5

Skill 3 — A minus in front of parentheses

Rule: $a - (b - c) = a - b + c$ — the minus applies to *every* term inside the parentheses, so each inside term flips its sign.

Example: Find $4 - (6 - 13)$

Step 1. A minus sign sits in front of a parenthesis, so distribute it to every term inside, flipping each sign, then combine left to right.

Step 2. $4 - (6 - 13) = 4 - 6 + 13 = 11$, combining the terms from left to right.

$$\Rightarrow 4 - (6 - 13) = \mathbf{11}$$

Try it: Find $2 - (5 - 9)$

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(!) Watch out: distributing the minus to only the *first* term inside is the most common sign error on this sheet. Flip every sign inside, or none.